

IMSD IN-CHARGE COMPLETE FIELD INSPECTION GUIDE

DFCCIL CIVIL ENGINEERING MAINTENANCE WING | DFC RAILROAD MANUAL (JULY-2024) AND ACS-01 TO ACS-05

Yeh guide **IMSD In-charge (Civil)** ke liye on-field inspection aur supervision ke samay ek complete reference handbook ke roop mein kaam karegi. Isme manual ke regular chapters ke sath-sath sabhi correction slips (**ACS-01, ACS-02, ACS-03, ACS-04, aur ACS-05**) ke exact check-points, measurements aur structural tolerances ko integrate kiya gaya hai. **Har ek Correction Slip update ko peeley rang (yellow fill) ke boxes mein highlight kiya gaya hai** taki aapko field par latest instructions ko refer karne mein koi paresani na ho.

SECTION 1: PLAIN TRACK & CURVES INSPECTION (■■■■■ ■■■■■■)

Jab aap plain track ya curves ka paidal (foot) ya trolley inspection kar rahe hon, toh niche likhe indicators aur values ko check karein:

1. Track Geometry & Tolerances (Floating Condition)

Parameter	Straight Track	Curves Radius ≥ 440m	Curves Radius < 440m
Track Gauge (Nominal 1676 mm)	-6 mm se +6 mm	-6 mm se +15 mm	Up to +20 mm allowed
Sleeper-to-Sleeper Gauge Var.	Max 2 mm allowed	Max 2 mm allowed	Max 2 mm allowed
Cross Level	±3 mm (Har 4th sleeper par check karein)	±3 mm	±3 mm
Alignment (10m Chord)	±2 mm deviation allowed	N/A	N/A
Twist (TW-1) on 3m base	PML/NBML: < 5 mm/m; UML: ≥ 7 mm/m (requires Speed Restriction)	N/A	N/A
Versine (20m Chord)	N/A	Var < 5 mm (for R > 600m)	Var < 10 mm (for R < 600m)

2. Physical & Structural Checkpoints

- **Zero Missing Fittings:** Pura track (fastenings, ERCs, liners, rubber pads) fully tight aur completed hona chahiye. Koi bhi missing fitting tolerate nahi ki jayegi.
- **Sleepers Spacing:** Theoretical spacing se variation ±20 mm se zyada na ho. Sleepers square hone chahiye.
- **Expansion Gaps (Non-LWR):** 20 successive gaps ka average gap check karein, tolerance ±2 mm honi chahiye. High joints max +2 mm tak allow hain.
- **Ballast Profile:** Shoulder aur crib ballast fully boxed aur compacted hona chahiye. Tamping machines ke liye min 150 mm clean cushion compulsory hai.

- **Curves Special Board:** Har curve ke dono tangent points par outside border par board check karein jo radius, length, transition length aur max superelevation show kare.
- **Gauge Face Lubricators:** Sharp curves par automatic gauge face lubricators ka operational status check karein (compulsory for curves $\geq 2.0^\circ$).

SECTION 2: POINTS & CROSSINGS / TURNOUTS (■■■■■■■ ■■■■■■■■■■■)

Turnouts DFCCIL track ka sabse critical aur heavy-load zone hote hain. Isme inspect karte samay niche likhe parameters ko strictly monitor karein:

1. Thick Web Switch (TWS) Assembly Checkpoints

- **SRJ to ATS Distance:** Stock Rail Joint (SRJ) se Actual Toe of Switch (ATS) ki distance **745.5 ± 6 mm** check karein.
- **ATS to JOH Distance:** ATS se Junction of Heads (JOH - jahan tongue rail width 72 mm ho jati hai) distance check karein: **8000 mm** (Straight Switch) aur **8013 mm** (Curved Switch).
- **Switch Throw:** ATS par point machine chalne par throw **160 mm** hona chahiye.
- **Closed Tongue Rail Housing:** Closed position mein tongue rail aur stock rail ke beech drive rod axis par gap ≤ **1 mm** hona chahiye.
- **JOH Clearance (Open Side):** Open side par tongue rail aur stock rail gap ≥ **60 mm** hona chahiye.
- **Slide Chairs & Rollers (Pi-Rollers):** Switch rollers perfectly clean aur working hon. Slide blocks aur tongue rail web gap ≤ **1 mm** ho.
- **Anti-Creep Device (ACD) Gaps:** Switch blade aur stock rail ke beech fitted ACD (pin & fork) ka ideal gap **7 mm at td** check karein. Agar ACD touch ho raha ho, toh use turant remove karwayein, nahi toh tongue rail ke peeche dangerous kink ban sakta hai.

■■ ADDENDUM & CORRECTION SLIP UPDATE - ACS-05 (DEC 2025) (TONGUE RAIL WEAR LIMITS)

Para 836(b)(i)(1) Switch Assembly updates: Tongue rail tab chipped/damaged mani jayegi jab continuous 10 mm length ke upar uski wear depth **10 mm se zyada** ho chuki ho (pehle yeh limit 8 mm thi). **Condemning Limit:** Agar chipped/cracked portion ki total length aggregate aur small pieces ko milakar **200 mm** se zyada ho jaye (within 1000 mm from ATS for 1 in 8.5, and within 2000 mm from ATS for 1 in 12), toh use badlein ya welding reconditioning ke liye mark karein. Chipped tongue rails ko reject karne ke bajay **welding process** dwara reclaim kiya ja sakta hai.

2. Lead Assembly & Crossing Assembly Checkpoints

Parameter	Standard Value / Maintenance Range
Crossing portion Track Gauge	Nominal 1676 mm. Strict tolerance -3 mm se +1 mm tak hi ho (baki turnout zone mein -3 mm se +3 mm).
Check Rail Clearance (Nose opposite)	Uniform clearance of 44 mm se 48 mm ke beech hona chahiye.
Check Rail Clearance (Flared ends)	Flared ends (end se 100 mm andar) par clearance 66 mm se 76 mm ke beech ho.
Check Rail Height (Raising)	Check rail running rail ke top se 20 mm upar (raised) honi chahiye.
Crossing Wear Limits	Wear < 6 mm: Safe. Wear 6-8 mm: Medium (schedule welding reconditioning). Wear > 8 mm: Immediate replacement.

■■ ADDENDUM & CORRECTION SLIP UPDATE - ACS-04 (OCT 2025) (GEOMETRIC MEASUREMENTS & ROBOTIC WELDING)

Para 835(b) & 846 updates: Points & Crossings detailed inspection ke dauran Gauge, Cross level aur Versine (turnout side) ke measurements **har 3-3 meter ke interval** par ATS (0 station) se lekar crossing ke back leg tak compulsory lein (previously irregular interval check). **Robotic Reconditioning:** Weldable CMS crossings ke in-situ reconditioning ke liye computer-controlled **Robotic Welding Technology** ka upyog karein. Par dhyān rahe, crossings jinke internal defects hon ya crack depth condemning size se **3 mm se zyada deep** ho, unhe robotic welding ke liye select na karein.

SECTION 3: SEJS (SWITCH EXPANSION JOINTS) & BRIDGES (■■■■■ ■■■■■■■■■■)

1. Switch Expansion Joints (SEJs)

- **Gaps Monitoring:** Reference mark aur tongue rail tip ke beech ka gap record karein. Standard movement design sheet ke gap limits se **±10 mm** ke andar hona chahiye.
- **Physical Integrity:** Har 15 din mein SEJ ki physical cleaning, oiling, greasing aur packing verify karein. Thande aur garam dino mein gap movement par kadi nazar rakhein.

2. Bridge Track Maintenance

- **Girder Seating:** Girders ki seating flush check karein. Koi bhi lateral ya longitudinal shift track alignment kharab karega.
- **Sleeper Fittings & Hook Bolts:** Steel channel sleepers par lagne wale hook bolts aur fittings ki tightening **har 15 din mein** (IMSD Sectional ke dwara) check aur tighten karna ensure karein.
- **Bridge Approaches:** Abutment ke paas immediate dip (rail level drop) nahi hona chahiye. Bridge abutment ke **3 m ke andar koi bhi rail joint** nahi hona chahiye.

SECTION 4: PATROLLING & EXTREME WEATHER CHECKS

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■■ ADDENDUM & CORRECTION SLIP UPDATE - ACS-02 (AUG 2025) (EXTREME WEATHER PATROLLING TRIGGERS)

Extreme Temperature Patrolling Limits: Extreme weather patrolling ke triggers aur beat rules ko badla gaya hai:

1. Hot Weather Patrolling: Jab actual rail temperature **td + 23°C** par pahunche tab patrolling shuru karwayein (pehle td + 25°C limit thi). Beat length: Single Line par **2 km**, Double Line par **1 km**.

2. Cold Weather Patrolling: Jab actual rail temperature **td - 28°C** par pahunche tab patrolling shuru karwayein (pehle td - 30°C limit thi). Beat length: Single Line par **4 km**, Double Line par **2 km**.

Safety Note: Sabhi patrolmen ke paas **GPS tracking device** compulsory provide karein aur real-time mapping monitor karein.

Hot Weather Action Checklist (Buckling/Kink Prevention)

Step 1: Immediate Train Halt: Kink ya incipient buckle notice hone par track ko standard distance par immediately protect karein.

Step 2: Floating Sleepers Inspection: Kink spot ke dono taraf 100-100 sleepers ka floating condition check karein.

Step 3: No Slew Rule: Track ko alignment me slew karne ya hilaane ki koshish bilkul na karein. Shoulders aur crib par additional ballast compact karein.

Step 4: Cooling the Rail: Rail temperature note karein aur rail head ko direct sun exposure se bachane ke liye use mitti, ballast ya gile joot (bags) se cover karein.

Monsoon & Special Structures

- **Bridge Waterways:** Pre-monsoon cleanup check karein aur HWM (High Water Mark) lines paint karwayein.
- **Tunnels (Para 266 & 123):** Pre-monsoon aur monsoon ke dauran **har mahine (monthly)** inspect karein. Standard clearance and inside-tunnel guard rails alignment verify karein.

SECTION 5: EMERGENCY ACTION & JERK MESSAGE PROTOCOL

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■■ ADDENDUM & CORRECTION SLIP UPDATE - ACS-03 (SEP 2025) (JERK MESSAGE ACTIONS)

Para 748 Jerk Message Updates: Control ya Station Master se jerk or unusual track message milte hi sabse tez conveyance se site par proceed karein. Affected location ke **dono taraf 500-500 meter (total 1 km track length)** ka complete foot inspection karein. Agar defect milta hai, toh immediate speed restriction lagakar rectify karwayein. **Night /**

Poor Visibility Clause: Agar night ke dauran defect ka accurate cause detect nahi ho pata aur visibility low hai, toh safety ke liye strictly **30 Kmph ya usse kam ■■ Speed Restriction (SR)** apply karein. Daylight (sunrise) ke baad mandatory re-inspection karwayein.

Rail / Weld Fracture Safety Protocols

Repair Methodology (Para 816/817)	Permissible Speed Rules
Standard Emergency Repair	Fish plates and clamps apply karke pehli train Stop Dead & 10 Kmph , aur subsequent trains ko 20 Kmph par pass karein.
High-Performance Rail Clamps	Under continuous watch (24 hrs watchman) pehli train Stop Dead & 10 Kmph , aur subsequent trains ko 50 Kmph par run hone dein.

SECTION 6: ADMINISTRATIVE, TOOLS & SAFETY CHECKS

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■■ ADDENDUM & CORRECTION SLIP UPDATE - ACS-01 (JUN 2025) (DEAD END INSPECTIONS)

Yard lines & Dead End inspections: Dead Ends / Buffer Stops ka inspection frequency update ki gayi hai. Pehle manual mein yeh saal mein ek baar (Once in a year) tha. Ab **ACS-01** ke anusaar IMSD In-charge aur Sectional ko rotation mein **har mahine (Once in a Month)** sabhi siding aur yard dead ends ka inspect karna compulsory hai.

- **Detonators Testing (Para 245):** Har saal apne stock aur outsourced staff ke stock se oldest detonator lekar empty 8-wheeled wagon ke niche safe range par walking speed par check karein. Normal life 5 years hai, satisfactory test se maximum 8 years tak extend ki ja sakti hai.
- **Trolleys Security (Para 1006):** Efficient tested brakes ke bina trolley track par na rakhein. Carry safety kit of 10 detonators, flags, and tri-colour rechargeable LED lanterns.
- **Land Boundary & SSOD:** March ending se pehle survey karke 'No Encroachment' aur SSOD structural monitoring verify karke 'No Infringement' certificate IMD In-charge ko submit karein.

Safety is our operating system. Let's maintain zero-accident tracks.